

Interpret Regression Coefficient Estimates

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0.1 Level-level Regression

$$y = \beta_0 + \beta_1 x + \epsilon$$

$$\frac{\partial y}{\partial x} = \beta_1$$

This is to say, if x rises by one unit, y increases by one unit.

0.2 Log-level Regression

$$\ln(y) = \beta_0 + \beta_1 x + \epsilon$$

Using the chain rule on the left hand side:

$$\frac{\partial \ln(y)}{\partial x} = \frac{1}{y} * \frac{\partial y}{\partial x} = \frac{\frac{\partial y}{\partial x}}{y}$$

This means that $\frac{\partial \ln(y)}{\partial x}$ is the change in y divided by the level of y . If we multiply by 100, we get the percentage change.

$$\frac{\partial \beta_0 + \beta_1 x + \epsilon}{\partial x} = \beta_1$$

$$\frac{\frac{\partial y}{\partial x}}{y} = \beta_1$$

$$\frac{\frac{\partial y}{\partial x}}{y} * 100 = \beta_1 * 100$$

So if you take β_1 , multiply it by 100, a one unit increase in x gives you a $\beta_1 * 100$ percentage change in y .

0.3 Level-log Regression

$$y = \beta_0 + \beta_1 \ln(x) + \epsilon$$

If we increase x by one percent we expect y to increase by $\beta_1/100$ units of y .

0.4 Level-level Regression

$$\ln(y) = \beta_0 + \beta_1 \ln(x) + \epsilon$$